Memo

Date: August 9, 2018
To: Mayor Donald M. Hahn
From: Jordan R. Kruse
Subject: Recommendation Report for State College Bicycle Lane Study

Attached to this letter is my report "An Analysis of Bicycle Infrastructure in State College, Pennsylvania." As described in the proposal, all research and studies were conducted pertaining to the addition of bicycle lanes in State College and what course of action the borough should take regarding them. This report aims to provide the borough with the information needed to benefit the health, safety, recreation, and transportation needs of its residents.

To provide this recommendation primary research within the area was conducted. This research consisted of observation of existing infrastructure along with a six-question survey of 68 students, faculty, and residents that bike in State College. Additional secondary research was conducted as a literature review of existing academic studies regarding bicycling and bike lanes. This primary and secondary research was analyzed based on four decision making criteria. These criteria consisted of safety, cost, health/environment, and travel efficiency. The relative importance of these criteria was then established in relation to each other, and a final recommendation was reached.

Results from the survey suggest that many residents of the State College area see a need for additional bicycle infrastructure in order to feel safer and make the area more bicycle friendly. The review of academic research concluded that bicycle lanes increase safety, come at a modest cost, provide a good long-term economic investment, benefit local health and environments, and improve travel ease and time for all road users. These results led to the recommendation that State College should invest in bicycle lanes in the immediate downtown area on its two busiest streets, College Ave. and Beaver Ave, for the estimated cost of $332,925.

Your interest and involvement in this recommendation report are greatly appreciated, and I would be very open to working with you again in the future. If you have any questions or comments, I can be reached at jxk5760@psu.edu or at (413) 835-5459.

Sincerely,

Jordan Kruse

Photograph from: http://www.pedbikeinfo.org/community/tips_bicyclist.cfm

AN ANALYSIS OF BICYCLE INFRASTRUCTURE IN STATE COLLEGE, PENNSYLVANIA

Abstract

This study analyzes the feasibility of constructing additional bicycle infrastructure in State College, Pennsylvania. In order to reach a recommendation, a six-question survey of local staff, students, and residents is taken into consideration. Along with this, four decision making criteria consisting of safety, cost, health/environment, and travel efficiency are analyzed using existing literature on bicycle infrastructure. These criteria are then weighted based on their importance and a final recommendation is reached.

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Glossary

Bicycle Lane – A section of a roadway marked exclusively for bicycle travel.

Bicycle Path – Any off-street pathway where bicycle travel is permitted.

Sharrows – Roadways with marks indicating bicycle travel, no dedicated lanes present.

Executive Summary

The borough of State College and the surrounding area is a very busy and dense area. Bicycles make up one of the primary modes of transport in the area, yet many people feel that the current infrastructure does not support cycling as much as it should. The aim of this study is to determine whether the borough of State College, PA should create more bicycle infrastructure and to make a recommendation on the topic.

Research for this study consisted of both primary and secondary research. Primary research was conducted in the form of a survey. This survey consisted of six questions and was answered by 68 respondents reached through various Penn State related online forums. The goal of this study was to understand public opinion on existing infrastructure, and to gauge whether people really feel if there is a need for more infrastructure. This study concluded that while people recognize the use of existing infrastructure, many residents feel that it is not enough and would support more.

Secondary research consisted of an in-depth literature review of four criteria that were to be analyzed being safety, cost, health/environment, and travel efficiency. Literature related to the safety of bike lanes provided evidence that bike lanes reduce accidents and injuries for cyclists. A review of cost estimated the cost of installing bike lanes to be around $133,170 per mile and showed that bike lanes give back to the economy in the form of economic opportunity and development, along with increasing property value. A review of health and environmental effects showed that bike lanes improve overall health and reduce CO2 production through the fact that they encourage more travel through active transport. Finally, academic literature showed that bicycle lanes improved travel efficiency by reducing slow downs caused by drivers passing cyclists.

These criteria were then weighted by their importance. Through analysis of the criteria, it was determined that the order of importance was safety as most important, followed by cost, health/environment, and travel efficiency in their respective orders. A recommendation of whether or not to install bike lanes was then made for each criterion individually, and then a final recommendation was chosen based on this. Almost all of the criteria fully supported the addition of bike lanes, with the only questionable category being cost.

On this basis, I recommend that State College implements bike lanes in its dense and busy downtown region. I believe that this will heavily benefit the health and safety of local residents and improve traffic efficiency while only costing a modest amount of money.

Introduction

This project's purpose is to conduct research and gather information to provide the borough of State College, Pennsylvania with a recommendation on whether to pursue creating additional dedicated bike lanes along many of its busier streets. Currently, cycling is one of the most popular methods of transportation in the State College area, yet only a small number of streets in the downtown State College area contain bike lanes or other dedicated cycling infrastructure. This bicyclist traffic without proper infrastructure both endangers currents cyclists and discourages others from cycling [1].

This report will analyze biking infrastructure in downtown State College, primarily the areas within 4 blocks of University Park, as this contains the densest areas of student housing and commercial space and sees some of the heaviest pedestrian, vehicle, and bike traffic. This recommendation will focus on bike lanes as a whole and will not distinguish between traditional and protected or buffered bike lanes. Along with this, existing sharrows will not be considered, as these are both inexpensive and generally have little effect of bicyclists and vehicle traffic [2], [3].

The methods used to reach a recommendation included in person observations of existing infrastructure, a survey of current users of State College bicycle infrastructure, and a review and analysis of existing literature regarding bike lanes. In person observations were conducted to establish background information regarding current infrastructure in downtown State College. A survey of 68 students, faculty, staff, and alumni was also conducted to understand perceived safety of and need for bicycle lanes. Finally, various criteria were analyzed using existing literature in order to reach a recommendation. These criteria consisted of safety, cost, health/environment, and travel efficiency.

The information gathered from this study and literature review has indicated that bicycle lanes increase safety, are relatively inexpensive, are a sound economic investment, benefit public health and the local environment, and improve traffic flow and efficiency. This information has led to a recommendation that the borough of State College, Pennsylvania create addition bicycle infrastructure in its dense downtown region.

Survey

In order to best get an idea of current public perception of both the current biking infrastructure at State College and University Park and the perceived need for new infrastructure, a survey was designed and presented to current students and users of State College biking infrastructure. This survey was completed by 68 respondents whom were reached using various Penn State related social media groups, primarily Reddit and Facebook.

Questions and Methodology

Questions for this survey were separated into 3 categories of two questions each, with the goals of establishing who the respondent is, understanding the respondent's perception of current safety conditions for cyclists in the area, and understanding the respondent's perceived need for additional bicycling infrastructure.

Table 1 - Questions and Reasonings

Question	Reasoning
What is your affiliation with Penn State University Park?	To better understand who the respondent is.
How often do you bike to or around campus?	To understand how relevant cycling is to the respondent's lifestyle.
How bicycle friendly do you feel the State College/University Park area is?	To establish how the respondent feels about the current ease of travel when cycling.
Do you feel safe when biking in or around the university?	To establish how the respondent feels about current safety conditions.
How do you feel about the current number of bicycle lanes and pathways?	To understand how the respondent feels about the amount of current infrastructure.
Do you think it would be beneficial to have more dedicated bicycle lanes and pathways?	To understand if the respondent feels there is a need for more infrastructure.

Results

68 respondents completed this survey. Of these respondents, 70.6% (N=48) were students, 14.7% (N=10) were local residents, and 14.7% (N=10) were university staff or alumni. Along with this, 45.5% (N=31) reported cycling to or around campus at least once per day, 25% (N=17) a few times per week, 14.7% (N=10) a few times per month, and 14.7% (N=10) reported never biking to or around campus.

When asked how bicycle friendly the State College area is, respondents reported an average rating of 3.16, indicating that respondents feel that while the State College area is not necessarily unfriendly to cyclists it isn't the most bicycle friendly either. When asked whether they feel safe bicycling in or around the university, 54% of respondents reported sometimes, 34% reported yes, and 12% reported no. This again shows a very average sense of safety, with the majority of respondents indicating that they only sometimes feel safe cycling in the area.

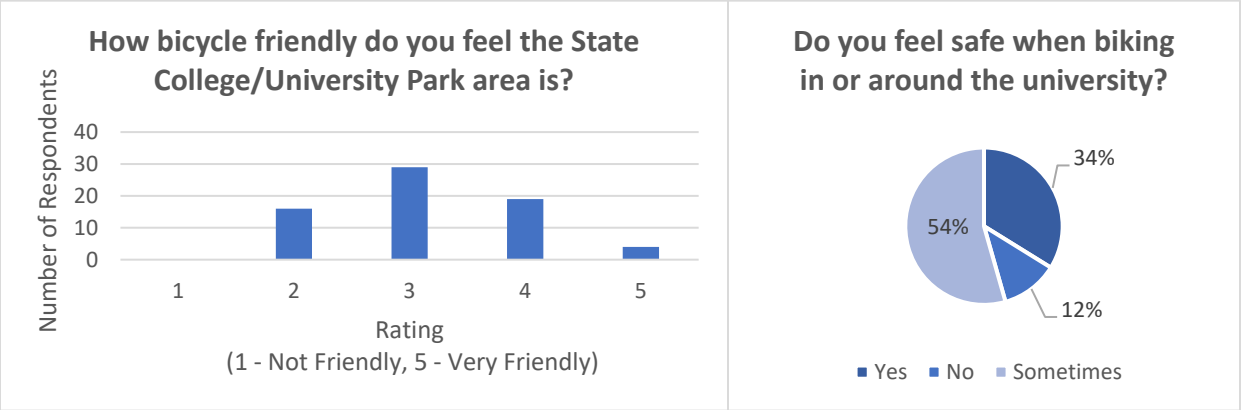

Figure 1 - Q3 & Q4: Bicycle Friendliness and Safety

When asked how they felt about the current number of bicycle lanes and pathways, the majority respondents indicated that the amount is decent, but still needs improvement. The next most popular answer was that the amount of bicycle lanes and pathways is not adequate. This clearly indicates that respondents feel there is a need for more bicycle lanes and pathways in the State College area. The final question asked was whether respondents feel if there should be more bicycle lanes and pathways. 87% of respondents answered "Yes" to this, with only 10% answering "No" and the remaining 3% responding with "No Opinion". While this clearly indicated that respondents feel that there is a need for more infrastructure, it should be noted that the phrasing of this question may have lead respondents to prefer a "Yes" response, and actual opinions may not be as extreme as indicated.

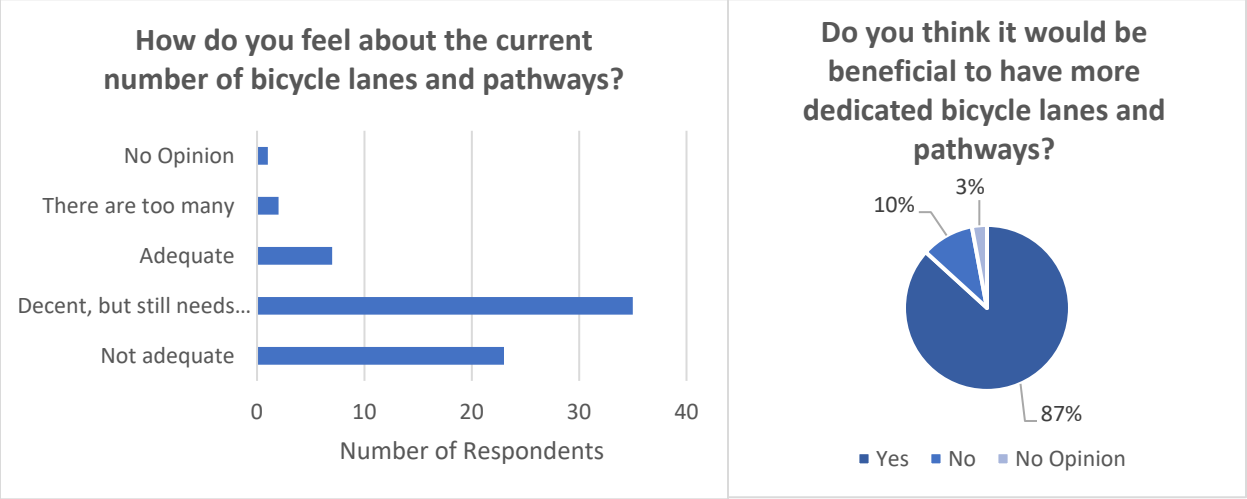

Figure 2 – Q5 & Q6: Bicycle Infrastructure Opinions

Survey Conclusion

Results of this survey indicated two primary themes:

1. While perceived safety and friendliness of current infrastructure is average, safety and cyclists' comfort still have much room for improvement.

2. Many respondents feel that current infrastructure is inadequate or still needs improvement and feel that the State College area would benefit from additional infrastructure.

The creation of bicycle lanes as a method to increase perceived safety by cyclists is supported by another study conducted on roadways in Michigan, which found that both cyclists and non-cyclists preferred greater roadway separation due to safety concerns [1].

From the results of this survey, it can be concluded that there is a lack of comfort with the current cycling infrastructure and public opinion supports the creation the of new infrastructure. This support for increased biking infrastructure indicates that such infrastructure would be appreciated and utilized, two factors that will have significant effects in the cost and health/environment criteria for decision making.

Criteria for decision making

In order to properly analyze the most import impacts of creating additional bike lanes, 4 important criteria were chosen that this decision will ultimately be based on. These criteria are safety, cost, health/environment, and traffic efficiency. Safety was chosen to be one of these criteria due to the fact that travel safety is paramount to the building of any infrastructure, whether it is for vehicles, pedestrians, or cyclists. Cost was chosen as the initial cost along with both long term maintenance and economic stimulation can have major impacts on any town reliant on a tight budget. Health and environment were both chosen as they have broader long-term impacts on both the residents of State College and its environment. These two sections are grouped as one due to the fact that they are both impacted by cycling usership, an aspect that will be discussed further on in this report. Finally, traffic efficiency was chosen, as cycling is primarily used as a mode of transport in State College, therefore it is important that it both makes travel easier for the cyclists and travelers in other modes of transport.

Safety

Safety is one of the largest issues with modern day transportation. In the United States in 2014 alone, there were 32,675 fatalities from traffic accidents, with 726 of these being cyclists [4]. The ability of bike lanes to reduce the injuries and fatalities for cyclists is controversial in public opinion, but literature supports the ability of bike lanes to reduce these accidents on crowded and busy streets, such as the ones seen in downtown State College.

A study conducted in Charlotte, North Carolina found that cyclists are at 3 to 4 times of an increased risk when on streets without bicycling lanes as opposed to those with bicycling lanes. This study took data from 36 streets with on street bicycle lanes and 26 streets without bicycle lanes and compared their accident rates. It was found that bicycle accidents occurred between 3 and 4 times as often on streets without bicycle lanes compared to those with bicycle lanes. Streets without bicycle lanes also saw between 140% to 180% increase in all crashes, suggesting bicycle lanes may also have an impact on motor vehicle safety. It was also concluded that the effectiveness of cycling lanes appeared to be dependent on traffic flow, meaning that bicycle lanes are more effective on busy streets [5].

Overall, this data shows that there is a clear correlation between the presence of on street bicycle lanes and a reduction in cycling accidents. Not only is greater separation between cyclists and drivers preferred by both groups, but this increased separation leads to a reduction in both cycling accidents and vehicle accidents, making roads safer for all.

Cost

While there is an easily quantifiable initial cost to the implementation of bike lanes, there are much more complex effects as a result of increased economic activity due to the increased cycling traffic that bike lanes often bring.

One study that used an aggregate of bicycle lanes bids found that the average cost of implement bicycle lanes came to $133,170 per mile [2]. The overall length of both College Ave. and Beaver Ave. in downtown state college is around 2.5 miles. As a result, creating bike lanes on these roads would come to a total cost of around $332,925. Along with this, maintenance of

on road bicycle lanes is often minimal due to the fact that it can be fixed alongside existing roadway infrastructure.

With the cost of bicycle lanes comes additional economic benefits. An analysis done by the Victoria Transport Policy Institute found that the addition of bicycling lanes can have positive effects on property values, economic opportunity, economic development. Property values of land near bicycle lanes often sees an increase, as these lanes are seen as a way to provide easy access between work, commercial areas, and homes. Along with this, the addition of bicycle lanes provides economic opportunity to those who may not be able to afford alternate transportation such as a car. Bicycle lanes can make longer distance job opportunities more accessible to those who have no option except to use active transport. Finally, bicycle lanes offer increased economic development, as the additional infrastructure provides consumers easier access to commercial areas [6].

Increased numbers of cyclists not only benefit the local economy but also save residents additional money. It is estimated that vehicle costs alone come to around around $2000 to $3000 per year, and vehicle operation costs come to about 10 to 15 cents per mile, with an even higher cost in congested areas and for short trips [6]. These costs are drastically reduced for cyclists, as bicycles generally cost between $100-$300 for a decent bike, and only cost around $50 a year for upkeep and maintenance, saving the user a considerable sum of money over time.

Overall, the addition of bicycle lanes comes at a relatively modest cost for the economic benefits provided. An implementation cost of around $330,000 to modify existing downtown roadways for bicycle lanes is a very effective investment. A study done on bike lanes in New York City concluded that bike lanes have a cost effectiveness ratio of around $1297/QALY[1], an extremely effective investment when it comes to improving quality of life [7]. The reduced consumer costs, additional economic opportunity, and increased economic development provide an overall good long-term investment for a growing town.

Health/Environment

Transport currently has a significant effect on both our health and the health of our environment. A case study in San Francisco found that the creation of bike lanes leads to an increase use of these lanes for travel [3]. Another study by the National Association of City Transport Officials found that as cities add more bike lanes, the number of overall cyclists also increases [8]. An increase in cyclists has the potential to improve public health from increased physical activity, while also reducing vehicle CO_2 emissions from a reduction in motor vehicle use.

On major area that bike lanes would provoke increased cycling from are from short trips, such as a few blocks down to a friend's apartment or to a store. An increase in cycling for these small trips rather than driving has very quantifiable health benefits. A study on health benefits of reduced car travel found that if all of these small trips by car were eliminated for a study area of 30 million people, 1295 deaths per year could be avoided [9]. Interpolated to

[1] Quality Adjusted Life Years

State College's population of 100,000, this has the potential to avoid approximately 4.3 deaths per year.

Along with this, increasing cycling ridership has the potential to drastically reduce the amount of CO_2 put into the atmosphere by vehicle emissions. A study on increasing the federal investment of bicycle riding in America found that through a modest shift to cycling which could reasonably be brought on from additional cycling infrastructure, the United States could see a yearly reduction of 21 million tons of CO_2 put into the atmosphere [10]. Interpolated to State College's population, this is a reduction of around 7,000 tons of CO2 per year in a modest scenario.

Overall, this increase in cycling ridership would result in significant health and environmental benefits for the State College area. While a complete elimination of short car trips due to bike lanes is not a reasonable expectation, the potential to avoid 4.3 deaths per year due to this shows a meaningful impact on the health of the overall population from an increase in cycling. Along with this, while the reduction in green space caused by additional bike lanes downtown is negligible due to existing infrastructure already existing and bike lanes having a rather small footprint, reducing State College's CO_2 emissions by 7,000 tons per year is very possible through the increase in cycling and could help State College be a leader in climate sustainability efforts.

Traffic Efficiency

Adding bike lanes to State College's roads has the potential to make travel easier for both cyclists and those traveling by motor vehicle. Bike lanes reduce the need for cars to slow down and swerve when passing cyclists. Along with this, studies have shown that adding bike lanes to roads also has positive effects of traffic flow.

A study conducted by Purdue University found that when motor vehicles have to pass cyclists on narrow or busy roads, they are required to both slow down and displace their vehicle to avoid the cyclist [11]. This study also found that even on wider roads, while the vehicle does not have to be displaced to comfortably avoid the cyclists, the vehicles still slow down. These slow downs due to cyclists can cause major traffic disruptions in busy conditions due to the nature of human drivers. Bike lanes eliminate this need for slowdowns and displacements, as the cyclists are further separated from drivers, and drivers are not taken by surprise when seeing cyclists in bike lanes as they might be when encountering them on unmarked roadways.

It is shown that the addition of bicycle lanes has improvements for all road vehicles. The presence of bike lanes removes the need of vehicles to swerve and slow down, reducing traffic jams as a result of this on busy streets. Along with this, traffic flow for both cyclists and motor vehicles sees and overall improvement due to the widened roads as a result of bicycle lanes [11]. These reduced travel times as a result of bike lanes have the ability to make transportation in State College both faster and easier.

Conclusion

As cycling grows in popularity it is important to properly analyze the infrastructure that it needs, along with the effects that implementing such infrastructure may have. Analyzing the

need for this infrastructure can be based on four main criteria: safety, cost, health/environment, and travel efficiency. In order to make a proper recommendation on this topic, it is important to weigh and compare these criteria properly in relation to each other.

The criteria that is deemed most important in this analysis is safety. The reasoning for this is that the ensuring safety and wellbeing of its citizens is a governments top priority, and that does not change when considering infrastructure. The next criteria of importance are the costs. This is due to the limited budget of State College and its need to allocate resources and money sparingly. The criteria of third-most importance are health and environmental concerns. While these are both long term considerations, they are both issues that are still of great important and have come to the forefront of public policy discussion, making them worthy of heavy consideration. The least important criteria listed is travel efficiency. While this is still an important consideration, it focuses more on convenience rather than the necessities of health and safety or the unchangeable realty of government budgets.

Table 2 - Criteria Analysis

Criteria	Analysis	Result
Safety	Bicycle lanes increase both actual and perceived safety for cyclists.	Implement Bicycle Lanes
Cost	Bicycle lanes come at a decent cost but offer long term economic benefits.	Possibly Implement Bicycle Lanes
Health/Environment	Bicycle lanes result in an increase in cycling, resulting in better public health and reduced vehicle pollution.	Implement Bicycle Lanes
Travel Efficiency	Bicycle lanes increase travel ease and traffic flow for all vehicles by reduced needs for motor vehicles to slow down or swerve to pass cyclists.	Implement Bicycle Lanes

Recommendation

My recommendation to the borough of State College is to create bike lanes by expanding existing roadways in the immediate downtown area, primarily along busy dense routes such as College Ave. and Beaver Ave.

This recommendation is based on the weighting of the 4 criteria as listed above. The most important criteria listed is safety. Through literature and public opinion, I determined that bike lanes significantly improve both the perceived and actual safety of cyclists, reducing accidents, injuries, and fatalities on routes with bike lanes versus those without. This increase in safety is shown to be far more effective on busier streets as opposed to quieter ones. The second criteria considered was cost, which was found to have an initial negative impact but provide long term benefits. While this initial cost can be harrowing, especially with tight town

budgets, it is far outweighed by the safety benefits and long-term economic benefits providing a potential positive return on investment. Although health and environmental considerations were less important, they still showed positive impacts in the long run, improving public health and working to prevent global climate change. Finally, even though travel efficiency was of least consideration its impact was still positive as bicycle lanes can eliminate traffic jams caused by cyclists being passed on roadways.

It is these overwhelmingly positive outcomes for criteria that have lead to be give a recommendation to create bike lanes in downtown State College. Although initial cost, an important factor, is a major issue to overcome, it is far outweighed by the positive results that the addition of bike lanes provides. These bike lanes would be most effective in the busiest areas of State College, making Beaver Ave. and College Ave. in downtown State College an optimal location for their implementation.

References

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Appendix A: Survey

Bike Lanes at PSU

We are looking at the current state of PSU bicycle infrastructure and whether people think it needs improvement. Please respond truthfully about your opinions on Penn State University Park and it's bicycle infrastructure. Thank you for your time!

* Required

1. **What is your affiliation with Penn State University Park?**
   Mark only one oval.

   - Student
   - Faculty
   - Other University Staff
   - Local Resident
   - Other: _____

2. **How often do you bike to or around campus?** *
   Mark only one oval.

   - Multiple times per day
   - Once per day
   - A few times per week
   - Once a week
   - A few times per month
   - Never

3. **How bicycle friendly do you feel the State College/University Park area is?** *
   Mark only one oval.

	1	2	3	4	5	
Not Bicycle Friendly						Very Bicycle Friendly

4. **Do you feel safe when biking in or around the university?** *
   Mark only one oval.

   - Yes
   - No
   - Sometimes

5. **How do you feel about the current number of bicycle lanes and pathways?** *

Mark only one oval.

- Not adequate
- Decent, but still needs improvement
- Adequate
- There are too many
- No Opinion

6. **Do you think it would be beneficial to have more dedicated bicycle lanes and pathways?** *

Mark only one oval.

- Yes
- No
- No Opinion

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